



中国认可
国际互认
检测
TESTING
CNAS L0095

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No.: RZUN2024-3950

检测报告

TEST REPORT

UN38.3

NAME OF SAMPLE:

Lithium and Iron Disulfide Battery

产品名称:

锂铁电池

CLIENT:

Guangzhou Great Power Energy Technology Co.,
Ltd.

委托单位:

广州鹏辉能源科技股份有限公司

CLASSIFICATION OF TEST:

Commission Test

检测类别:

委托测试

威凯检测技术有限公司
CVC Testing Technology Co., Ltd.

检测报告

TEST REPORT

No.: RZUN2024-3950

Page 2 of 13 Pages

Name of samples: Lithium and Iron Disulfide Battery 样品名称: 锂铁电池	Type/Model: 型号规格: LFB-AA, FR6 1.5V 3500mAh
Color: Black 样品颜色: 黑色	Physical shape: Cylindrical 样品形状: 圆柱形
Commissioned by: Guangzhou Great Power Energy Technology Co., Ltd. 委托单位: 广州鹏辉能源科技股份有限公司	Commissioner address: No.912, West Village Segment, Shiliang Road, Shawan Town, Panyu District, Guangzhou City, Guangdong Province 511483, P.R. China 委托单位地址: 广州市番禺区沙湾镇市良路(西村段) 912 号
Manufacturer: Guangzhou Great Power Energy Technology Co., Ltd. 制造商: 广州鹏辉能源科技股份有限公司	Manufacturer address: No.912, West Village Segment, Shiliang Road, Shawan Town, Panyu District, Guangzhou City, Guangdong Province 511483, P.R. China 制造商地址: 广州市番禺区沙湾镇市良路(西村段) 912 号
Factory: Guangzhou Great Power Energy Technology Co., Ltd. 生产厂: 广州鹏辉能源科技股份有限公司	Factory address: No.912, West Village Segment, Shiliang Road, Shawan Town, Panyu District, Guangzhou City, Guangdong Province 511483, P.R. China 生产厂地址: 广州市番禺区沙湾镇市良路(西村段) 912 号
Classification of test: Commission Test 检测类别: 委托测试	Quantity of sample: 40 cells 样品数量: 40 个电芯
Tested according to: 测试标准: ST/SG/AC.10/11/Rev.8/ Section 38.3	Sample identification: 样品标识序号: c1#~c40#
Receiving date: 接样日期: 2024-06-12	Means of receiving: Submitted by commissioner 接样方式: 委托单位送样
Completing date: 完成日期: 2024-07-03	Test item: 7 items 测试项目: 7 项
Test conclusion: 检测结论: The Lithium and Iron Disulfide Batteries submitted by Guangzhou Great Power Energy Technology Co., Ltd. are tested according to Section 38.3 of the Eighth revised edition of the Manual of Tests and Criteria (ST/SG/AC.10/11/Rev.8/Section 38.3). The test items are full items. The test results comply with the relevant requirements of the standard. 由广州鹏辉能源科技股份有限公司送检的锂铁电池, 依据联合国《试验和标准手册》第八修订版第 38.3 节进行检测, 试验为全项目, 试验结果符合标准相关要求。 Seal of CVC CVC 盖章 Date of issue: 签发日期: 2024-07-10	

Title: Manager
批准人职务: 经理

Approved by: Huang Kun

Reviewed by: Zhang Siyao

Tested by: Zheng Feida

批 准:

Huang Kun

审 核:

Zhang Siyao

检 测:

Zheng Feida

Description and illustration of the sample:

样品说明及描述:

The sample's status is good

样品状况良好。

Cell Dimensions/电芯尺寸: $\phi 14\text{mm} \times 50\text{mm}$

Aggregate Lithium content of each cell/ 单个电芯的锂含量: 0,98g

Test item 试验项目	Sample No. 样品编号	State 状态	Remark 备注
T.1~T.5	c1#~c10#	in undischarged states 未放电状态	-
	c11#~c20#	in fully discharged states 完全放电状态	-
T.6	c21#~c25#	in undischarged states 未放电状态	-
	c26#~c30#	in fully discharged states 完全放电状态	-
T.8	c31#~c40#	in fully discharged states 完全放电状态	-

Description of the sampling procedure:

取样程序的说明:

/

Description of the deviation from the standard, if any:

试验结果不符合标准项的说明:

/

Remarks:

备注:

Throughout this report a comma is used as the decimal separator.

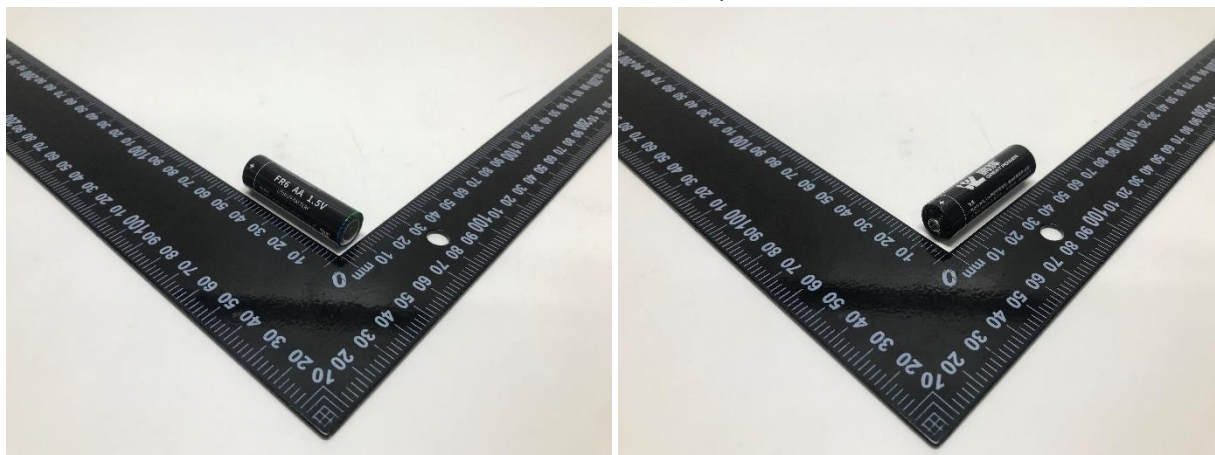
本报告中以逗号代替小数点。

The samples that have models of LFB-AA, FR6 are the same battery, only different in colors and labels.

该样品型号有 LFB-AA, FR6 区别仅在于颜色及标签不同, 其它完全一致, 均属于同一款电池。

Photos of Samples and Labels/样品照片及标识

Cell/电芯 (LFB-AA, FR6 1.5V 3500mAh)



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ST/SG/AC.10/11/Rev.8/ Section 38.3			
Clause 章节	Requirements 标准要求	Result 测试结果	Verdict 判定
38.3.4	Procedure/试验步骤		—
38.3.4.1	Test T.1: Altitude simulation/试验 T.1: 高度模拟		P
	Test cells and batteries shall be stored at a pressure of 11,6kPa or less for at least six hours at ambient temperature (20±5℃)/ 将电芯和电池在温度为 20±5℃，大气压力为不大于 11,6kpa 的环境中贮存不少于 6 个小时		
	Requirement/标准要求: 1 Cells and batteries Mass loss limit: ≤0,2% /样品质量损失≤0,2% 2 Open circuit voltage not less than 90%, The requirement relating to voltage is not applicable to test cells and batteries at full discharged states. 样品试验后开路电压应不低于试验前开路电压的 90 %,此要求不适用于完全放完电的电池和电芯。 3 No leakage, no venting, no disassembly, no rupture and no fire 样品（电池）应无漏液、无排气、无解体、无破裂以及无着火现象的发生	The samples c1#~c20# : No leakage, no venting, no disassembly, no rupture and no fire/编号为 c1#~c20#的样品：无漏液、无排气、无解体、无破裂以及无着火现象 The data is shown in Table 1./数据见表 1	
38.3.4.2	Test T.2: Thermal test/试验 T.2: 温度试验		P
	Test cells and batteries are to be stored for/电池存储条件如下: 1 For small cells and batteries: one temperature cycle: 72±2℃(6h) ——40±2℃(6h) / 对于小型电芯和电池：一次温度循环为 72±2℃(6h) ——40±2℃(6h) For large cells and batteries: one temperature cycle: 72±2℃(12h) ——40±2℃(12h) / 对于大型电芯和电池：一次温度循环为 72±2℃(12h) ——40±2℃(12h) 2 The maximum time interval between test temperature extremes is 30 minutes/温度转换最大间隔时间为 30min 3 This procedure is to be repeated 10 times/重复 10 次循环 4 after which all test cells and batteries are to be stored for 24 hours at ambient temperature (20±5℃)/循环结束后，电池在 20±5℃的条件下 搁置 24 小时		
	Requirements/标准要求 1 Cells and batteries Mass loss limit: ≤0,2% /样品质量损失≤0,2% 2 Open circuit voltage not less than 90%, The requirement relating to voltage is not applicable to test cells and batteries at full discharged states. 样品试验后开路电压应不低于试验前开路电压的 90 %,此要求不适用于完全放完电的电池和电芯。 3 No leakage, no venting, no disassembly, no rupture and no fire 样品（电池）应无漏液、无排气、无解体、无破裂以及无着火现象的发生	The samples c1#~c20# : No leakage, no venting, no disassembly, no rupture and no fire/编号为 c1#~c20#的样品：无漏液、无排气、无解体、无破裂以及无着火现象 The data is shown in Table 1./数据见表 1	

ST/SG/AC.10/11/Rev.8/ Section 38.3			
Clause 章节	Requirements 标准要求	Result 测试结果	Verdict 判定
38.3.4.3	Test T.3: Vibration/试验 T.3: 振动 1 Cells and batteries are firmly secured to the platform of the vibration machine / 电芯和电池牢固地安装在振动台（的台面）上 2 The vibration: a sinusoidal waveform with a logarithmic sweep between 7Hz and 200Hz and back to 7Hz traversed in 15 minutes/振动以正弦波形式，以 7Hz 增加至 200Hz，然后在减少回到 7Hz 为一个循环，一个循环持续 15 分钟的对数前移传送。 3 For cells and small batteries: from 7 Hz a peak acceleration of $1g_n$ is maintained until 18Hz is reached. The amplitude is then maintained at 0,8mm (1,6mm total excursion) and the frequency increased until a peak acceleration of $8g_n$ occurs (approximately 50Hz). A peak acceleration of $8g_n$ is then maintained until the frequency is increased to 200Hz. / 对于电芯和小型电池：从 7Hz 开始，以 $1g_n$ 的峰值加速度保持不变，直到达到 18Hz。然后将振幅保持在 0,8mm（总偏移 1,6mm）并且频率增加直到出现 $8g_n$ 的峰值加速度（大约 50Hz）。然后保持 $8g_n$ 的峰值加速度，直到频率增加到 200Hz。 For large batteries: from 7Hz a peak acceleration of $1g_n$ is maintained until 18Hz is reached. The amplitude is then maintained at 0,8mm (1,6mm total excursion) and the frequency increased until a peak acceleration of $2g_n$ occurs (approximately 25Hz). A peak acceleration of $2g_n$ is then maintained until the frequency is increased to 200Hz. / 对于大型电池：从 7Hz 开始，以 $1g_n$ 的峰值加速度保持不变，直到达到 18Hz。然后将振幅保持在 0,8mm（总偏移 1,6mm）并且频率增加直到出现 $2g_n$ 的峰值加速度（大约 25Hz）。然后保持 $2g_n$ 的峰值加速度，直到频率增加到 200Hz。 4 This cycle repeated 12 times for a total of 3 hours for each of three mutually perpendicular mounting position of the cell. One of the directions of vibration must be perpendicular to the terminal face. / 以振动的其中一个方向必须是垂直样品极性，对每个电芯从三个互相垂直的方向上循环 12 次，每个方向 3 个小时，共 9 小时。		P
	Requirements/标准要求 1 Cells and batteries Mass loss limit: $\leq 0,2\%$ / 样品质量损失 $\leq 0,2\%$ 2 Open circuit voltage not less than 90%, The requirement relating to voltage is not applicable to test cells and batteries at full discharged states. 样品试验后开路电压应不低于试验前开路电压的 90%，此要求不适用于完全放完电的电池和电芯。 3 No leakage, no venting, no disassembly, no rupture and no fire 样品（电池）应无漏液、无排气、无解体、无破裂以及无着火现象的发生	The samples c1#~c20# : No leakage, no venting, no disassembly, no rupture and no fire/编号为 c1#~c20# 的样品：无漏液、无排气、无解体、无破裂以及无着火现象 The data is shown in Table 1./数据见表 1	

ST/SG/AC.10/11/Rev.8/ Section 38.3			
Clause 章节	Requirements 标准要求	Result 测试结果	Verdict 判定
38.3.4.4	Test T.4: Shock/Test T.4: 冲击 1 Test cells and batteries shall be secured to the testing machine/以稳固的托架固定住每个电芯和电池样品的全部配件表面。 2 Each cell shall be subjected to a half-sine shock of peak acceleration of 150 g_n and pulse duration of 6 milliseconds. Large cells may be subjected to a half-sine shock of peak acceleration of 50 g_n and pulse duration of 11 milliseconds. / 对每个电芯以峰值为 150g_n 的半正弦的加速度撞击，脉冲持续 6 毫秒，大型电芯须经受最大加速度 50g_n 和脉冲持续时间 11 毫秒的半正弦波冲击。 Small batteries shall be subjected to a half-sine shock of peak acceleration of 150 g_n (or Acceleration(g_n)= $\sqrt{\left(\frac{100850}{mass}\right)}$, which is smaller) and pulse duration of 6 milliseconds, large batteries shall be subjected to a half-sine of peak acceleration of 50 g_n (or Acceleration(g_n)= $\sqrt{\left(\frac{30000}{mass}\right)}$, which is smaller) and pulse duration of 11 milliseconds/对每个电池以峰值为 150g_n (或与 $\sqrt{\left(\frac{100850}{mass}\right)}$ 中的较小值) 的半正弦的加速度撞击，脉冲持续 6 毫秒，大型电池须经受最大加速度 50g_n (或与 $\sqrt{\left(\frac{30000}{mass}\right)}$ 中的较小值) 和脉冲持续时间 11 毫秒的半正弦波冲击。 3 Each cell or battery shall be subjected to three shocks in the positive direction followed by three shocks in the negative direction of three mutually perpendicular mounting positions of the cell or battery for a total of 18 shocks/每个电池或电池组须在三个互相垂直的电池安装方位的正方向经受三次冲击，接着在反方向经受三次冲击，总共经受 18 次冲击。		P
	Requirements/标准要求: 1 Cells and batteries Mass loss limit: ≤0,2% /样品质量损失≤0,2% 2 Open circuit voltage not less than 90%, The requirement relating to voltage is not applicable to test cells and batteries at full discharged states. 样品试验后开路电压应不低于试验前开路电压的 90 % ,此要求不适用于完全放完电的电池和电芯。 3 No leakage, no venting, no disassembly, no rupture and no fire 样品（电池）应无漏液、无排气、无解体、无破裂以及无着火现象的发生	The samples c1#~c20# : Acceleration= 150g_n No leakage, no venting, no disassembly, no rupture and no fire/编号为 c1#~c20#的样品: 峰值加速度= 150g_n 无漏液、无排气、无解体、无破裂以及无着火现象 The data is shown in Table 1./数据见表 1	

ST/SG/AC.10/11/Rev.8/ Section 38.3			
Clause 章节	Requirements 标准要求	Result 测试结果	Verdict 判定
38.3.4.5	Test T.5: External Short Circuit/试验 T.5 外部短路		P
	1 The cell or battery to be tested shall be heated for a period of time necessary to reach a homogeneous stabilized temperature 57±4℃/加热电芯或电池样品直到温度稳定在 57±4℃ 2 The cell or battery shall be subjected to a short circuit condition with a total external resistance of less than 0,1 ohm at 57±4℃, This short circuit condition is continued for at least one hour after the cell or battery external case temperature has returned to 57±4℃, or in the case of the large batteries, has decreased by half of the maximum temperature increase observed during the test and remains below that value. /将样品正负极用小于 0,1Ω 的总电阻回路进行短路，样品的外表温度恢复到 57±4℃之后保持短路状态 1 小时以上；对于大电池，电池温度降低至最高温升值的一半时实验结束。 3 The cell or battery must be observed for a further six hours for the test to be concluded, /对电芯或电池必须进一步观察 6 个小时才能下结论。		
	Requirements/标准要求: During the test and within six hours after test ,the cells or batteries 在试验过程中以及之后 6 个小时内，电芯或电池样品 1. External temperature not exceed 170℃ 外表温度不超过 170℃ 2. No disassembly, no rupture and no fire. 无解体、无破裂和无着火现象发生。	The samples c1#~c20# : no disassembly, no rupture and no fire/编号为 c1#~c20# 的样品：无解体、无破裂以及无着火现象 The data is shown in Table 1./数据见表 1	

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ST/SG/AC.10/11/Rev.8/ Section 38.3			
Clause 章节	Requirements 标准要求	Result 测试结果	Verdict 判定
38.3.4.7	Test T.7: Overcharge/试验 T.7: 过度充电		N/A
	1 The charge current shall be twice the manufacturer's recommended maximum continuous charge current/以 2 倍制造厂推荐的最大持续充电电流对样品充电 2 The minimum voltage of the test shall be as follows/本试验最小电压见下文		
	a) When the manufacturer's recommended charge voltage is not more than 18V, the minimum voltage of the test shall be the lesser of two times the maximum charge voltage of the battery or 22V/ 如果厂家推荐的充电电压不超过 18V, 本试验的最小充电电压应是厂家标定最大充电电压的两倍或者是 22V 之中的较小者。 b) When the manufacturer's recommended charge voltage is more than 18V, the minimum voltage of the test shall be 1,2 times the maximum charge voltage/ 如果厂家推荐的充电电压超过 18V, 本试验的最小充电电压应是厂家标定最大充电电压的 1,2 倍。 3 Tests are to be conducted at ambient temperature 20±5℃, The duration of the test shall be 24 hours/20±5℃的环境温度下, 试验持续 24 小时。	—	
	Requirements/标准要求: No disassembly and no fire within seven days of this test 试验样品在试验中和试验后 7 天内, 应无解体和无着火现象发生。	—	
38.3.4.8	Test T.8: Forced discharge/试验 T.8: 强制放电		P
	Each cell shall be forced discharged at ambient temperature by connecting it in series with a 12 V D.C. power supply at an initial current equal to the maximum discharge current specified by the manufacturer, 20±5℃的环境温度下, 将单个电芯连接在 12V 的直流电源上进行强制放电, 此直流电源提供给每个电芯初始电流为制造厂指定的最大放电电流。		
	The specified discharge current is to be obtained by connecting a resistive load of the appropriate size and rating in series with the test cell. Each cell shall be forced discharged for a time interval (in hours) equal to its rated capacity divided by the initial test current (in ampere) 指定的放电电流通过串联在试验电芯上的合适大小和功率的负载来获得, 每个电芯的强制放电时间 (小时) 为额定容量除以初始电流 (安培)。		
	Requirements/标准要求: No disassembly and no fire within seven days of this test 试验样品在试验中和试验后 7 天内, 应无解体和无着火现象发生。	The samples c31#~c40#: No disassembly and no fire /编号为 c31#~c40#的样品: 无解体、无着火现象 For voltage data before test, see table 3. / 试验前电压见表 3	

Table1: T1~T5 / 表 1. 试验 1~试验 5											
Sample No. 样品号	Mass prior to test / 试验前质量 (g)	OCV prior to test / 试验前电压 (V)	Test T.1: Altitude simulation/ 试验 T.1: 高度模拟		Test T.2: Thermal test/ 试验 T.2: 温度试验		Test T.3: Vibration/ 试验 T.3: 振动		Test T.4: Shock/ 试验 T.4: 冲击		Test T.5: External Short Circuit/试验 T.5 外部短路
			Mass Loss(%) 质量损失(%)	OCV Retention Ratio(%) 电压保留比(%)	Mass Loss(%) 质量损失(%)	OCV Retention Ratio(%) 电压保留比(%)	Mass Loss(%) 质量损失(%)	OCV Retention Ratio(%) 电压保留比(%)	Mass Loss(%) 质量损失(%)	OCV Retention Ratio(%) 电压保留比(%)	Temp. (°C) 温度 (°C)
c1#	15,748	1,821	0,000	100,00	0,038	100,00	0,000	100,00	0,000	100,00	80,4
c2#	15,807	1,821	0,000	100,00	0,037	100,00	0,000	100,00	0,000	100,00	80,2
c3#	15,963	1,817	0,000	100,00	0,037	100,00	0,000	100,00	0,000	100,00	81,8
c4#	15,751	1,819	0,000	100,00	0,044	100,00	0,000	100,00	0,000	100,00	81,4
c5#	15,820	1,819	0,000	100,00	0,031	100,00	0,000	100,00	0,000	100,00	81,0
c6#	15,833	1,816	0,000	100,00	0,031	100,00	0,000	100,00	0,000	100,00	81,2
c7#	15,710	1,825	0,000	100,00	0,031	100,00	0,000	100,00	0,000	100,00	81,4
c8#	15,895	1,816	0,000	100,00	0,031	100,00	0,000	100,00	0,000	100,00	81,0
c9#	15,752	1,818	0,000	100,00	0,031	100,00	0,000	100,00	0,000	100,00	80,8
c10#	15,787	1,819	0,000	100,00	0,025	100,00	0,000	100,00	0,000	100,00	81,5
c11#	15,961	—	0,000	—	0,018	—	0,000	—	0,000	—	58,2
c12#	16,079	—	0,000	—	0,012	—	0,000	—	0,000	—	57,9
c13#	15,956	—	0,000	—	0,031	—	0,000	—	0,000	—	58,3
c14#	15,953	—	0,000	—	0,025	—	0,000	—	0,000	—	58,1
c15#	15,969	—	0,000	—	0,012	—	0,000	—	0,000	—	58,0
c16#	15,941	—	0,000	—	0,018	—	0,000	—	0,000	—	58,2
c17#	15,999	—	0,000	—	0,025	—	0,000	—	0,000	—	58,4
c18#	16,008	—	0,000	—	0,024	—	0,000	—	0,000	—	58,1
c19#	15,880	—	0,000	—	0,037	—	0,000	—	0,000	—	58,4
c20#	16,012	—	0,000	—	0,037	—	0,000	—	0,000	—	58,0

Table2: Crush 表 2:挤压											
Test T.6: Crush 试验 T.6:挤压	Sample No. 样品号	c21#	c22#	c23#	c24#	c25#	c26#	c27#	c28#	c29#	c30#
	OCV prior to test / 试验前电压 (V)	1,816	1,819	1,816	1,820	1,818	1,458	1,379	1,402	1,459	1,553
	Temp. (°C) 温度 (°C)	35,8	34,9	34,1	32,7	30,9	30,2	29,8	30,6	29,4	30,1

Table 3: Forced discharge / 表 3. 强制放电											
Test T.8: Forced discharge / 试验 T.8: 强制放电	Sample No. 样品号	c31#	c32#	c33#	c34#	c35#	c36#	c37#	c38#	c39#	c40#
	OCV prior to test / 试验前电压(V)	2,952	2,950	2,948	2,950	2,950	2,948	2,949	2,949	2,948	2,948

注 意 事 项

Important

1. 报告无检测单位印章无效。
The test report is invalid without the seal of CVC.
2. 未经本试验室书面同意，不得部分地复制本报告。
Nobody is allowed to photocopy or partly photocopy this test report without written permission of CVC.
3. 本报告无批准人、审核人及检测人签名无效。
The test report is invalid without the signatures of Ratifier, Reviewer and Testing engineer.
4. 本报告涂改无效。
The test report is invalid if altered,
5. 对检测报告若有异议，应于收到报告之日起十五天内向检测单位提出。
Objections to the test report must be submitted to CVC within 15 days.
6. 本报告仅对送检样品负责。
The test report is valid for the tested samples only.
7. 判定栏中“-”表示“不需要判定”，“P”表示“通过”，“F”表示“不通过”，“N/A”表示“不适用”。
As for the Verdict, “-” means “no need for judgement”, “P” means “pass”, “F” means “fail” and “N/A” means “not applicable”.

****报告中未加 CMA 标志时，检测数据和结果仅供科研、教学或内部质量控制之用。****
The test data and test results given in this test report should only be used for purposes of scientific research, teaching and internal quality control when the CMA symbol is not presented.

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